

This part below is wrong - it was when the CSV file for gdp_per_capita used the 2020 year instead of 2000 year!

This query yields the following output: total_rows as 434, nonnull_gdp as 211, nonnull_health as 377, nonnull_pop65 as 434, and nonnull_death as 434

Confused at why only getting 211 nonnull_gdp rows - so I used the following code to see if there was a difference in the years for which gdp was covered vs other variables (e.g.

```
pop_age_65_plus table)
SELECT MIN(year),MAX(year)
FROM pop_age_65_plus;
```

```
SELECT MIN(year),MAX(year)
FROM gdp_per_capita;
```

This code said min was 2019 and max was 2020

```
SELECT COUNT(*)
FROM pop_age_65_plus p
JOIN gdp_per_capita g
ON p.country_code = g.country_code
AND p.year = g.year;
```

This query directly checks for overlap - it yields 217 rows

So GDP non_null after joining was 211 rows, and direct overlap with INNER JOIN was 217 rows
The INNER JOIN includes rows where a match exists in both tables - BUT this includes rows where gdp_per_capita might still be NULL

Whereas the non_null_gdp after LEFT JOIN only includes rows where gdp exists AND gdp is NOT NULL!

```
SELECT COUNT(*)
FROM pop_age_65_plus p
JOIN gdp_per_capita g
ON p.country_code = g.country_code
AND p.year = g.year
WHERE g.gdp_per_capita IS NULL;
```

This query checks for number of rows where gdp_per_capita is null - and it yields 6 rows → so
 $217 - 211 = 6$ null rows

So missing values just inside the GDP table is 19 rows

Coverage mismatch (no match at all between values) shrinks dataset from 434 rows to 217 rows (INNER JOIN)

And there are 6 rows for the null values - these are the null values within the matched rows when we do the INNER JOIN

So missingness arises from incomplete dataset coverage and null values within matched observations (via the INNER JOIN)

So, to handle the data coverage mismatch, we will use the complete cases approach where we only use the observations that have valid values!

```
SELECT COUNT(*)  
FROM combined_data  
WHERE gdp_per_capita IS NOT NULL  
AND current_health_expenditure_per_capita IS NOT NULL;
```

This query tells us the true usable data - our final dataset size, where all the variables are NON NULL! → we didn't need to check the population age 65+ or the death rate data for NON NULL values because those are perfect datasets, and we already know all of their values are NON NULL → The query yielded 190 rows

So that means we have 190 usable observations out of 434 original observations → retained 44% of the original data

Lost rows because GDP had limited overlap, health had missing values, and combining led to the intersection shrinking to 190 rows

Now, I will create the final modeling view (i.e. with this cleaned dataset)

```
CREATE VIEW modeling_data AS  
SELECT *  
FROM combined_data  
WHERE gdp_per_capita IS NOT NULL  
AND current_health_expenditure_per_capita IS NOT NULL;
```

But the problem with this is that I only see values for the year 2019!

Question: Why not include all 3 predictors (health expenditure, gdp per capita, pop age 65+) for 2000 and 2019, and include just the rows where all 3 predictors have non-null values for both years?

Answer: So, including only rows where all 3 predictors are non-null for 2000 and 2019 means we have the same country for both years, and there are non-null predictors for both years. The

problem with this is that we could end up with USA in 2000 and India in 2019 if we used a SQL query like the one below:

```
WHERE year IN (2000, 2019)
AND gdp IS NOT NULL
AND health IS NOT NULL
AND pop IS NOT NULL
```

So, what I am really asking for is countries that have complete data in both 2000 AND 2019! In other words, I am looking to create a balanced panel/2-year panel.

SOLUTION:

Remaining steps we follow to get the 2 year panel

1. Identify Valid rows (non-null predictors)

```
CREATE VIEW valid_rows AS
SELECT *
FROM combined_data
WHERE year IN (2000,2019)
AND gdp_per_capita IS NOT NULL
AND current_health_expenditure_per_capita IS NOT NULL
and pop_65_plus IS NOT NULL;
```

BUT...this first step creates a problem because it only shows data values with the year 2019! It completely excludes 2000

This means that for the year 2000, at least 1 of the 3 predictors is NULL for every row, which causes the 2000 to get filtered out.

```
SELECT
year,
COUNT(*) AS total_rows,
COUNT(gdp_per_capita) AS nonnull_gdp,
COUNT(current_health_expenditure_per_capita) AS nonnull_health,
COUNT(pop_65_plus) AS nonnull_pop65
FROM combined_data
WHERE year IN (2000,2019)
GROUP BY year
ORDER by year;
```

Getting 0 nonnull rows for gdp_per_capita

	year numeric 🔒	total_rows bigint 🔒	nonnull_gdp bigint 🔒	nonnull_health bigint 🔒	nonnull_pop65 bigint 🔒
1	2000	217	0	184	217
2	2019	217	211	193	217

***I made a typo! In wb_gdp_per_capita CSV file I cleaned, I wrote 2020 instead of 2000 - so the GDP table had 2020 instead of 2000, so when I joined on (country_code, year), the 2000 rows from the other tables had no match in GDP, and GDP showed null for all the 2000 rows

So filtering gdp_per_capita IS NOT NULL removed all the 2000 rows!

**Data misalignment due to key inconsistency!

How to fix this:

First, I fixed the CSV → so I went to wb_gdp_per_capita CSV and replaced 2020 with the correct year of 2000. Then, I reloaded the table with the following query:

```
DROP TABLE gdp_per_capita
```

Then, I recreated the table and re-imported the corrected CSV file (with 2000 as the year now).

The problem with the above SQL query is that I can't drop the gdp_per_capita table since the other views I created (combined_data, modeling_data, valid_rows all depend on gdp_per_capita in some way, direct or indirectly)

Solution: Use the following SQL query instead:

```
DROP TABLE gdp_per_capita CASCADE;
```

This will remove the table and all dependent views (the combined_data, modeling_data, valid_rows)

So I used the CASCADE command, and that was step 2.

Now, for step 3: I will recreate the gdp_per_capita table, with the correct CSV file (using 2000 as the year) this time.

I reloaded the gdp_per_capita table now.

Step 4: Recreating the views.